Data-Backed Journal-Style Replication of *Overcoming Local Search Through Alliances and Mobility*

Replication Paper Project

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Abstract

This paper presents a methodology-specific local analogue of Overcoming Local Search Through Alliances and Mobility using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The paper is therefore reported as methodology-specific local analogue rather than as a source-data replay. The current local batch contains 23,901 observations across 184 entities, and the batch headline estimate is 0.0722 for Mobility Bridge x Technological Distance with p = 0.002. The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: local search; alliances; mobility; technological distance; patent knowledge flows; local calculated analogue

1 Introduction

This paper begins from a paper-first reading of Overcoming Local Search Through Alliances and Mobility and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The present manuscript is classified as methodology-specific local analogue. A paper-first restatement of the empirical question is as follows: The source paper asks how firms overcome

local search and reach distant knowledge through alliances and mobility (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication matters because the paper is a core statement of how firms escape localized search without abandoning their existing knowledge base entirely. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 23,901 observations across 184 entities and spans 1986 to 2016 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Rosenkopf and Almeida, 2003a; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent bridge-mechanism analogue that can be estimated from the current processed patent and citation-network stack. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Rosenkopf and Almeida, 2003a; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper sits in a top-journal literature on localized search, interfirm knowledge flows, and the mechanisms that help firms bridge distance. (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the local rerun is about mechanism transportability: whether the value of bridge mechanisms survives after the tie architecture is translated into processed network and recombination variables. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search,

technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003b; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). The paper-specific review therefore deepens the network and knowledge-flow conversation rather than relying only on generic exploration references. (Rosenkopf and Almeida, 2003b; Almeida and Kogut, 1999; Ahuja, 2000). A second topic-specific literature block is added to sharpen the manuscript's empirical boundary and identification problem. (Stuart and Podolny, 1996; Schilling and Phelps, 2007). A final topic-specific literature block keeps the bibliography tied to the paper's actual mechanism rather than to a generic replication template. (Fleming et al., 2007; Fleming and Sorenson, 2004; Lavie et al., 2011).

2 Search Distance, Bridge Mechanisms, and Replication Boundary

The focal source article is (Rosenkopf and Almeida, 2003a). The original study can be summarized as a mechanism paper with a sharp empirical core. The published paper studies alliances and inventor mobility as mechanisms that help firms overcome local search. Technological distance is a central moderator in the argument. The empirical outcome concerns knowledge flows and innovative usefulness rather than simple patent volume. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and

March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Original Paper Methodology

A faithful evaluation of Rosenkopf and Almeida (2003) starts with the published empirical design rather than with the local rerun. The source article first establishes geographic and technological localization of knowledge in the semiconductor industry and then evaluates whether alliances and the mobility of active inventors bridge otherwise distant knowledge contexts (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published sample is described as follows: The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting. The published measurement architecture is described as follows: The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows. These details matter because they determine what the published paper actually observes rather than what a later analogue can observe (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The published estimator is described as follows: patent-citation-based models of interfirm knowledge flows that relate those flows to alliances, inventor mobility, and technological distance in semiconductors. The critical published comparison is whether alliances and inventor mobility become more useful as technological distance rises, conditional on the original semiconductor patent-citation setting. Reading the source methodology this way prevents the current manuscript from quietly replacing the original design with one generic fixed-effects story (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.2 Replication Status and Boundary

Replication status for this paper: Methodology-specific local analogue (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). This paper is a methodology-specific local analogue because the original article directly observes alliances and inventor mobility in semiconductors, whereas the local rerun maps those mechanisms onto citation-network bridges and outside-the-firm search cases. The current data can test the bridge logic, but not replay the original mechanism archive one-for-one (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

2.3 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Alliances and mobility are the two bridge mechanisms. The local reconstruction therefore adopts the following bounded decision: Alliance bridges are proxied with rolling citation-network ties and mobility bridges with outside-the-firm search cases. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Technological distance moderates the value of those bridges. The local reconstruction therefore adopts the following bounded decision: Technological distance is mapped to recombination breadth in the local patent panel. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline

(Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The outcome is about knowledge-usefulness rather than raw output. The local reconstruction therefore adopts the following bounded decision: Patent-level inventive impact is used as the executable outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

3 Local Patent, Bridge, and Distance Data

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Patent-level recombination panel with organizational and field experience counts from the local USPTO focal-patent workflow. Rolling firm-year interfirm citation-link network measures built from processed backward patent ties. Patent-level inventive impact and patenting controls from the cleaned reproduction dataset (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 23,901 observations across 184 entities and covers 1986 to 2016 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly

instead of pretending that the original archive has been fully recreated (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high. The local sample frame is Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures The observed time window in the executable panel is 1986 to 2016 in the local executable panel The outcome is Patent-level inventive impact (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Rosenkopf and Almeida, 2003a; Hall et al., 2005).

3.1 Patent-Level Sample Reconstruction

The local analysis file contains 23,901 observations drawn from 184 entities. The observed time span runs from 1986 to 2016. Selected sample moments are Impact = 5.309, Alliance Bridge = 44.16, Technological Distance = 4.291 (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	23,901
Unique Firms	184
Start Year	1986
End Year	2016
Mean Impact	5.309
Mean Alliance Bridge	44.16

Statistic	Value
Share Mobility Bridge	13.74%
Mean Technological Distance	4.291

Sample reconstruction from the current local dataset.

3.2 Bridge, Distance, and Impact Measures

The construct Alliance bridge is operationalized locally as Rolling direct citation-network ties and appears in the executable dataset as alliance_bridge. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use direct ties from the five-year rolling interfirm citation-link network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Mobility bridge is operationalized locally as Outside-the-firm but within-field search indicator and appears in the executable dataset as mobility_bridge. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Classify patents with zero organizational but positive field experience as mobility-like bridge cases.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja,

2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Technological distance is operationalized locally as Breadth of recombination and appears in the executable dataset as technological_distance. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the recombination-breadth measure from the focal-patent panel.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Knowledge-flow usefulness is operationalized locally as Inventive impact and appears in the executable dataset as inventive_impact. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the local patent-level inventive-impact measure.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Patent controls is operationalized locally as Degree, patent stock, firm age and appears in the executable dataset as recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the already materialized patent-level controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters

the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Alliance bridge	Rolling direct citation-network ties	Use direct ties from the five-year rolling interfirm citation-link network.
Mobility bridge	Outside-the-firm but within-field search indicator	Classify patents with zero organizational but positive field experience as mobility-like bridge cases.
Technological distance	Breadth of recombination	Carry forward the recombination-breadth measure from the focal-patent panel.
Knowledge-flow usefulness	Inventive impact	Use the local patent-level inventive-impact measure.
Patent controls	Degree, patent stock, firm age	Reuse the already materialized patent-level controls.

Variable mapping and derivation logic in the main text.

3.3 Local Dataset Families Used in the Bridge Analogue

This replication uses the local patent-level recombination panel from the focal-patent workflow, a rolling firm-year citation-link network derived from backward patent ties, and the cleaned inventive-impact panel with patent-level controls (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). Those local dataset families are not interchangeable: patent-level recombination panel is used to measure technological distance, recombination, and mobility-like search states; rolling citation-link network is used to proxy alliance-like bridges at the firm-year level; cleaned inventive-impact panel is used to supply inventive impact and patent-level controls. Together, they support 23,901 observations across 184 entities over 1986 to 2016 in the local executable panel (Rosenkopf

and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). This local data stack is narrower than the original article's data architecture. Patent-level recombination panel with organizational and field experience counts from the local USPTO focal-patent workflow. Rolling firm-year interfirm citation-link network measures built from processed backward patent ties. Patent-level inventive impact and patenting controls from the cleaned reproduction dataset. The manuscript therefore keeps the boundary visible: the source paper supplies the original methodology, whereas the local processed dataset families and calculated outputs supply the estimable analogue. Exact filenames are retained in the replication notes rather than printed through the body text (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005).

DatasetFamily	RoleInPaper
Patent-level recombination panel	measure technological distance, recombination, and mobility-like search states
Rolling citation-link network	proxy alliance-like bridges at the firm-year level
Cleaned inventive-impact panel	supply inventive impact and patent-level controls

Local data families used in the present paper.

3.4 Construct Translation from Alliances and Mobility

The derivation appendix records how each major construct enters the executable estimand for Rosenkopf and Almeida (2003). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Alliance bridge, the manuscript uses Rolling direct citation-network ties under the exact local implementation alliance_bridge. The derivation step is: Use direct ties from the five-year rolling interfirm citation-link network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Mobility bridge, the manuscript uses Outside-the-firm but within-field search indicator under the exact local implementation mobility_bridge. The derivation step is: Classify patents with zero organizational

but positive field experience as mobility-like bridge cases.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technological distance, the manuscript uses Breadth of recombination under the exact local implementation technological_distance. The derivation step is: Carry forward the recombination-breadth measure from the focal-patent panel.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Knowledge-flow usefulness, the manuscript uses Inventive impact under the exact local implementation inventive_impact. The derivation step is: Use the local patent-level inventive-impact measure.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Patent controls, the manuscript uses Degree, patent stock, firm age under the exact local implementation recombination_degree; total_patents_same_year; total_patents_until_prev_log; firm_age. The derivation step is: Reuse the already materialized patent-level controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Bridge and Distance Patterns in the Local Patent Panel

For a local-search paper, the descriptive section has to show how often bridge mechanisms appear and how far patents are actually reaching across technological distance. For Inventive impact, the mean is 5.309, the standard deviation is 17.53, and the interquartile range runs from 1.000 to 6.000; For Alliance bridge, the mean is 44.16, the standard deviation is 29.95, and the interquartile range runs from 4.000 to 67.00; For Mobility bridge, the mean is 0.137, the standard deviation is 0.344, and the interquartile range runs from 0.00000 to 0.00000; For Technological distance, the mean is 4.291, the standard deviation is 4.563, and the interquartile range runs from 2.000 to 5.000 . The central descriptive question is whether alliance-like ties, mobility-like search, and technological distance vary enough to make the bridge-by-distance interactions interpretable rather than mechanical (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time profile is read as a history of search conditions.

Changes in bridge prevalence or average distance matter because the source theory predicts that bridges become especially valuable when firms face harder search problems. the value of patent observations is 646 in 1986 and 534 in 2016; the value of mean inventive impact is 4.015 in 1986 and 3.768 in 2016; the value of mean alliance-like bridge intensity is 0.00000 in 1986 and 0.00000 in 2016 . These temporal shifts determine whether the later regression is identifying the paper's mechanism in a stable panel or in one whose observable context changes meaningfully across the sample window (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The nonparametric support check is treated as a distance gradient. It asks whether inventive impact changes visibly as patents move from lower to higher observed technological distance. Mean outcomes range from 4.466 to 8.110 across the support of the focal regressor, while the mean regressor itself ranges from 2.411 to 9.961. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation matrix is interpreted as a bridge-mechanism check. If alliance-like and mobility-like measures were nearly the same object, the later interaction evidence would be much less persuasive. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Bridge Mechanisms and Distance Support

The figure is chosen to behave like a bridge-mechanism figure (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The left panel shows whether alliance-like and mobility-like bridges expand in the same periods as technological distance, and the right panel shows how inventive impact changes across the observed

distance gradient. mean alliance-like bridge intensity moves from 0.00000 in 1986 to 0.00000 in 2016; share of mobility-like bridge patents moves from 25.85% in 1986 to 4.49% in 2016; mean technological distance moves from 2.865 in 1986 to 4.629 in 2016 Across the support panel, mean outcome moves from 4.466 to 8.110 as the mean regressor shifts from 2.411 to 9.961 (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

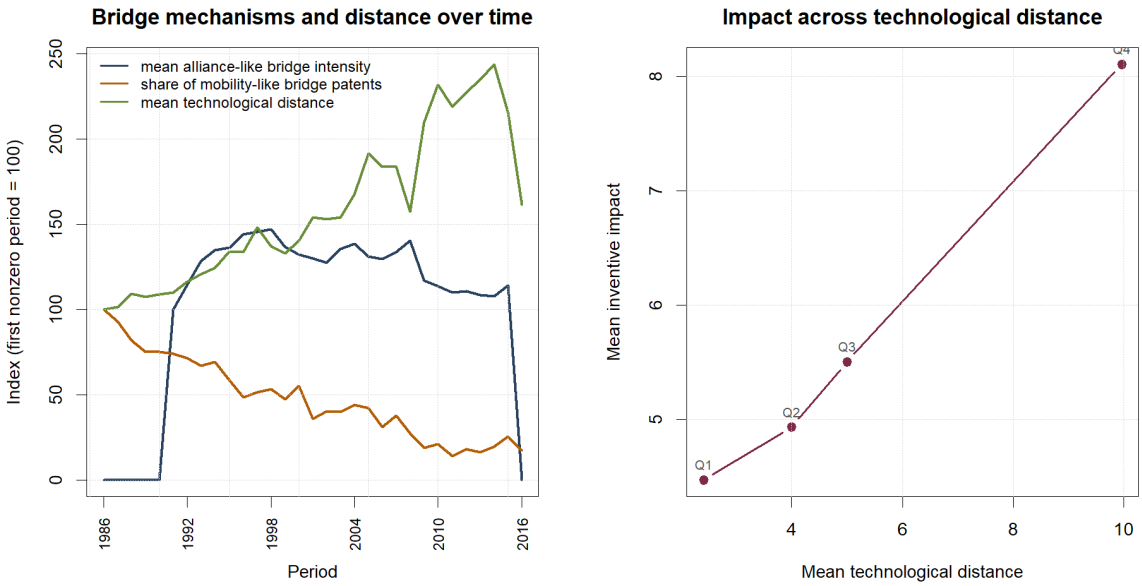

Figure 1: Figure 1. Bridge prevalence, technological distance, and the observed impact profile across the local search-distance support.

4.2 Bridge Coverage and Distance Support

For a local-search paper, data quality is about whether bridge variables and distance are measured with enough range to make the interaction design meaningful. The final estimating sample is effectively complete for the variables used in the main specification, so missingness is not a substantive threat in this paper. For that reason, the manuscript does not spend body-table space on an all-zero missingness display (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). Coverage is reported to show whether the bridge-by-distance result is learned from a broad set of patenting firms rather than from a narrow set of bridge-heavy organizations. The most heavily represented entities contribute between 830 and 3,200 observations each. That concentration check is reported

in prose because the relevant question is whether a few entities dominate the sample, not which firm names happen to be listed in a long table (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). Distance support is central because the source theory predicts that bridge mechanisms matter especially when search becomes harder. In the lowest observed quantile, the mean regressor is 2.411 and the mean outcome is 4.466 across 13,716 observations. In the highest observed quantile, the corresponding values are 9.961, 8.110, and 4,443 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). These diagnostics matter because a bridge paper can only support its mechanism if distance, bridge prevalence, and impact vary jointly in the data. The section therefore focuses on support where the interaction claim actually lives (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005).

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	4.466
Q2	3,465	4.000	4.930
Q3	2,277	5.000	5.503
Q4	4,443	9.961	8.110

Outcome support across the observed quantiles of the focal regressor.

Period	Patents	MeanImpact	MeanAllianceBridge	MobilityBridgeShare	MeanDistance
1986	646	4.015	0.0000	25.85%	2.865
1987	793	4.296	0.0000	24.09%	2.914
1988	1,071	4.469	0.0000	21.20%	3.135
1989	1,398	3.926	0.0000	19.46%	3.084
1990	1,360	4.507	0.0000	19.49%	3.117
1991	1,333	4.749	46.89	19.20%	3.151
1992	1,309	5.007	53.68	18.49%	3.329
1993	1,358	5.678	60.25	17.38%	3.465
1994	1,337	6.272	63.28	17.88%	3.572
1995	1,217	6.260	63.87	15.12%	3.844
1996	659	6.727	67.66	12.59%	3.838
1997	646	6.014	68.22	13.31%	4.245

Period	Patents	MeanImpact	MeanAllianceBridge	MobilityBridgeShare	MeanDistance
1998	356	6.295	68.92	13.76%	3.924
1999	654	5.483	64.08	12.23%	3.810
2000	636	5.541	62.11	14.31%	4.022
2001	614	5.151	60.93	9.28%	4.414
2002	632	4.889	59.78	10.44%	4.384
2003	522	4.502	63.52	10.34%	4.418
2004	324	3.664	65.01	11.42%	4.809
2005	284	4.500	61.54	10.92%	5.486
2006	336	4.890	60.79	8.04%	5.265
2007	246	3.654	62.65	9.76%	5.268
2008	535	3.652	65.82	7.10%	4.512
2009	206	4.801	54.93	4.85%	6.005
2010	511	4.417	53.42	5.48%	6.648
2011	651	6.452	51.58	3.69%	6.275
2012	1,068	4.000	51.94	4.68%	6.519
2013	868	5.386	50.90	4.26%	6.730
2014	725	12.64	50.58	5.10%	6.989
2015	1,072	6.616	53.53	6.62%	6.174
2016	534	3.768	0.0000	4.49%	4.629

Observed time patterns relevant to the focal empirical setting.

5 Empirical Strategy for Bridge Mechanisms

The estimand targeted in this manuscript is A patent-level firm and year fixed-effects model of inventive impact on alliance-like bridges, mobility-like bridges, technological distance, and their interactions. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit

of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Rosenkopf and Almeida, 2003a; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

5.1 Why the Local Bridge Proxies Are Used

The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures, and the empirical window is 1986 to 2016 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Patent-level inventive impact, and the main explanatory design is Patent-level fixed-effects model relating inventive impact to alliance-like bridges, mobility-like bridges, technological distance, and the interaction of each bridge with distance. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Recombination degree, same-year patenting, prior patent stock, firm age, and firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps patents with verified recombination and impact variables and merges only to firm-year network observations supported by the rolling citation-link graph.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf

and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. For this paper, the status is methodology-specific local analogue. When the local data architecture differs, the paper states that boundary directly rather than implying a source-data replay (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures over 1986 to 2016 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Patent-level inventive impact and the main explanatory design is Patent-level fixed-effects model relating inventive impact to alliance-like bridges, mobility-like bridges, technological distance, and the interaction of each bridge with distance. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Recombination degree, same-year patenting, prior patent stock, firm age, and firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps patents with verified recombination and impact variables and merges only to firm-year network observations supported by the rolling citation-link graph. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

5.2 Patent-Level Estimating Equation

$$\log(1+\text{Impact}_{pft}) = \beta_1\text{AllianceBridge}_{ft} + \beta_2\text{MobilityBridge}_p + \beta_3\text{Distance}_p + \beta_4\text{AllianceBridge}_{ft} \times \text{Distance}_p + \beta_5\text{MobilityBridge}_p \times \text{Distance}_p + \beta_6 X_{pft} + \alpha_f + \gamma_t + \varepsilon_{pft}. \tag{1}$$

6 Bridge Mechanisms and Search-Distance Results

6.1 Main Bridge-by-Distance Estimates

The central local estimate is 0.0722 on Mobility Bridge x Technological Distance (p = 0.002). The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. Technological distance is estimated at 0.0533 with p = ¡ .001. Recombination degree is estimated at -0.478 with p = ¡ .001. Mobility bridge is estimated at -0.225 with p = ¡ .001 (Rosenkopf and Almeida, 2003a; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Alliance bridge	-0.0008	0.0020	0.702
Mobility bridge	-0.2247	0.0596	0.0007
Technological distance	0.0533	0.0091	2.03e-06
Recombination degree	-0.4784	0.0838	3.16e-06
Same-year patents	-4.41e-05	2.13e-05	0.0476
Log prior patent stock	-0.1036	0.0702	0.150
Alliance bridge x technological distance	-0.0002	0.0002	0.274
Mobility bridge x technological distance	0.0722	0.0215	0.0022

Key coefficient table used in the main text.

6.2 Interpreting the Distance Interactions

The coefficient pattern is not being summarized as a bare sign check. Technological distance is estimated at 0.0533 with a p-value of ¡ .001; Recombination degree is estimated at -0.478 with a p-value of ¡ .001; Mobility bridge is estimated at -0.225 with a p-value of ¡ .001; Mobility bridge x technological distance is estimated at 0.0722 with a p-value of 0.002 (Rosenkopf and Almeida,

2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Technological distance, which is estimated at 0.0533 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Inventive impact has mean 5.309 and standard deviation 17.53 in the local panel, while the main regressor Technological distance has mean 4.291 and standard deviation 4.563 (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 4.466 to 8.110 while the mean regressor ranges from 2.411 to 9.961. This does not substitute for the model, but it shows how much raw variation the specification must explain (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The executable analogue emphasizes whether bridge mechanisms become more valuable as recombination distance increases, so the alliance-by-distance and mobility-by-distance interactions are central. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Evidence on Bridge Mechanisms

The coefficient figure is especially useful here because the paper's central claim is about the relative role of alliance-like bridges, mobility-like bridges, and their interactions with distance (Rosenkopf and Almeida, 2003a; Hall et al., 2005).

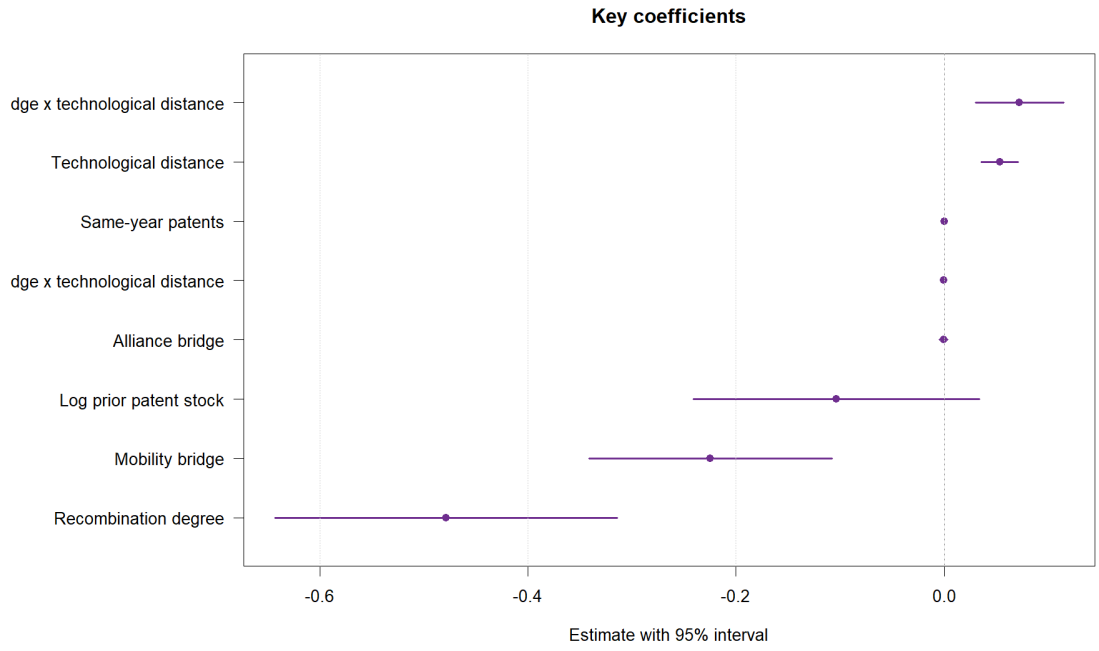

Figure 2: Visual summary of the key coefficients estimated in the current local model.

6.4 Diagnostic Checks

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). For local_search_bridges, the current fit export reports N = 23,884, pseudo R-squared = 0.0508, R-squared = 0.132 (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Rosenkopf and Almeida (2003) is being carried out on a clearly described estimating sample with observable fit statistics (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005).

6.5 Alternative Model Readings

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Technological distance with estimate 0.0533 and p = ¡ .001 (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988;

Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Technological distance is estimated at 0.0533 with p = ¡ .001 and is classified as a strong signal; Recombination degree is estimated at -0.478 with p = ¡ .001 and is classified as a strong signal; Mobility bridge is estimated at -0.225 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Rosenkopf and Almeida (2003) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

6.6 Supporting Coefficient Families

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Technological distance = 0.0533 (p = ¡ .001); Recombination degree = -0.478 (p = ¡ .001); Mobility bridge = -0.225 (p = ¡ .001); Mobility bridge x technological distance = 0.0722 (p = 0.002); Same-year patents = -4.41e-05 (p = 0.048). The largest positive estimate in this specification is Mobility bridge x technological distance at 0.0722, while the largest negative estimate is Recombination degree at -0.478. Reading the specification this way matters because it shows which part of the model is doing the explanatory

work and which part is mainly serving as adjustment structure (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Rosenkopf and Almeida (2003) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Rosenkopf and Almeida, 2003a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

7 Comparison with the Published Alliance-and-Mobility Study

7.1 Published Bridge Logic Versus the Local Patent Panel

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Rosenkopf and Almeida, 2003a). For Rosenkopf and Almeida (2003), the local headline result is Mobility Bridge x Technological Distance = 0.0722 with p = 0.002. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper finds that alliances and mobility help firms access distant knowledge and that their usefulness grows with technological distance.	Mobility Bridge x Technological Distance = 0.0722 with p = 0.002.	The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact.
Measurement and data construction	The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows.	The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.

Aspect	OriginalPaper	LocalDataset	Reading
Sample and unit of analysis	The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting.	Patent-level recombination panel merged to rolling firm-year interfirm citation-network measures. Local sample = 23,901 observations across 184 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Implications of the Distance-Sensitive Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design uses direct alliance and inventor-mobility evidence together with semiconductor patent citation patterns to trace interfirm knowledge flows By contrast, the local paper relies on the following executable design: The local analogue maps alliances to rolling interfirm citation-link ties and mobility to patents that use knowledge new to the focal firm but not entirely new to the field, then asks whether these bridges are more useful when technological distance is high These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published sample follows a cohort of semiconductor firms and studies interfirm knowledge flows in that industry setting The local sample instead works from Patent-level recombination panel merged to rolling firm-year interfirm

citation-network measures and yields 23,901 observations across 184 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. The present manuscript is classified as methodology-specific local analogue. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988;

Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The published paper argues that both alliances and mobility become more useful as technological distance rises. The local analogue asks whether a citation-network bridge and an outside-the-firm search bridge display the same distance-sensitive pattern in patent-level inventive impact (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Rosenkopf and Almeida, 2003a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source paper observes alliances and inventor mobility directly in semiconductor data (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun uses alliance-like citation-network ties and a mobility-like outside-the-firm search indicator because direct inventor mobility and alliance archives are not part of the current executable stack (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The paper therefore evaluates the bridge logic in a transparent analogue rather than claiming a line-by-line replay of the original data architecture (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent bridge-mechanism analogue that can be estimated from the current processed patent and citation-network stack (Rosenkopf and Almeida, 2003a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001;

Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Rosenkopf and Almeida, 2003a; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	23,901	rows
Unique Firms	184	unique_firms
Start Year	1986	year_min
End Year	2016	year_max
Mean Impact	5.309	mean_impact
Mean Alliance Bridge	44.16	mean_alliance_bridge
Share Mobility Bridge	13.74%	share_mobility_bridge
Mean Technological Distance	4.291	mean_technological_distance

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Inventive impact	23,901	5.309	17.53	1.000	2.000	6.000	0	812
Alliance bridge	23,901	44.16	29.95	4.000	51.00	67.00	0	99
Mobility bridge	23,901	0.137	0.344	0.0000	0.0000	0.0000	0	1
Technological distance	23,901	4.291	4.563	2.000	3.000	5.000	2	117
Recombination degree	23,901	0.555	0.174	0.444	0.560	0.691	0	1
Same-year patents	23,901	727.1	1,302	109.0	311.0	652.0	1	7321
Log prior patent stock	23,901	7.500	2.217	6.324	7.684	8.893	0	11
Firm age	23,901	13.62	9.194	6.000	10.00	22.00	1	31

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Inventive impact	Alliance bridge	Mobility bridge	Technological distance	Recombination degree	Same-year patents	Log prior patent stock	Firm age
Inventive impact	1.000	0.033	-0.017	0.364	0.018	0	0.022	0.023
Alliance bridge	0.033	1.000	-0.100	0.094	0.067	0	0.619	0.385
Mobility bridge	-0.017	-0.100	1.000	-0.138	-0.144	0	-0.180	-0.179
Technological distance	0.364	0.094	-0.138	1.000	0.466	0	0.149	0.260
Recombination degree	0.018	0.067	-0.144	0.466	1.000	0	0.122	0.156
Same-year patents	-0.018	0.338	-0.087	0.011	0.065	1	0.520	0.323
Log prior patent stock	0.022	0.619	-0.180	0.149	0.122	1	1.000	0.743
Firm age	0.023	0.385	-0.179	0.260	0.156	0	0.743	1.000

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanImpact	MeanAllianceBridge	MobilityBridgeShare	MeanDistance
1986	646	4.015	0.0000	25.85%	2.865
1987	793	4.296	0.0000	24.09%	2.914
1988	1,071	4.469	0.0000	21.20%	3.135
1989	1,398	3.926	0.0000	19.46%	3.084
1990	1,360	4.507	0.0000	19.49%	3.117
1991	1,333	4.749	46.89	19.20%	3.151
1992	1,309	5.007	53.68	18.49%	3.329
1993	1,358	5.678	60.25	17.38%	3.465
1994	1,337	6.272	63.28	17.88%	3.572
1995	1,217	6.260	63.87	15.12%	3.844
1996	659	6.727	67.66	12.59%	3.838
1997	646	6.014	68.22	13.31%	4.245
1998	356	6.295	68.92	13.76%	3.924
1999	654	5.483	64.08	12.23%	3.810
2000	636	5.541	62.11	14.31%	4.022
2001	614	5.151	60.93	9.28%	4.414
2002	632	4.889	59.78	10.44%	4.384
2003	522	4.502	63.52	10.34%	4.418
2004	324	3.664	65.01	11.42%	4.809
2005	284	4.500	61.54	10.92%	5.486
2006	336	4.890	60.79	8.04%	5.265
2007	246	3.654	62.65	9.76%	5.268
2008	535	3.652	65.82	7.10%	4.512
2009	206	4.801	54.93	4.85%	6.005
2010	511	4.417	53.42	5.48%	6.648
2011	651	6.452	51.58	3.69%	6.275
2012	1,068	4.000	51.94	4.68%	6.519
2013	868	5.386	50.90	4.26%	6.730

Period	Patents	MeanImpact	MeanAllianceBridge	MobilityBridgeShare	MeanDistance
2014	725	12.64	50.58	5.10%	6.989
2015	1,072	6.616	53.53	6.62%	6.174
2016	534	3.768	0.0000	4.49%	4.629

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	13,716	2.411	4.466
Q2	3,465	4.000	4.930
Q3	2,277	5.000	5.503
Q4	4,443	9.961	8.110

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
alliance_bridge	-0.0008	0.0020	-0.387	0.70
mobility_bridge	-0.2247	0.0596	-3.774	¡ .001
technological_distance	0.0533	0.0091	5.867	¡ .001
recombination_degree	-0.4784	0.0838	-5.708	¡ .001
total_patents_same_year	-4.41e-05	2.13e-05	-2.066	0.048
total_patents_until_prev_log	-0.1036	0.0702	-1.476	0.15
alliance_bridge:technological_distance	-0.0002	0.0002	-1.115	0.27
mobility_bridge:technological_distance	0.0722	0.0215	3.351	0.002

Full coefficient dump exported from the local estimation script.

model	n	logLik	AIC	BIC	pseudo_r2	r2	within_r2
local_search_bridges	23,884	-31,492	63,394	65,051	0.0508	0.132	0.0332

Model fit statistics from the local estimation output.

References

Ahuja, G. (2000). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. and Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3):197–220.

Ahuja, G. and Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6-7):521–543.

Almeida, P. and Kogut, B. (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science*, 45(7):905–917.

Anderson, P. and Tushman, M. L. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3):439–465.

Benner, M. J. and Tushman, M. L. (2003). Exploitation, exploration, and process management: The productivity dilemma revisited. *Academy of Management Review*, 28(2):238–256.

Cohen, W. M. and Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132.

Fleming, L., Mingo, S., and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, 52(3):443–475.

Fleming, L. and Sorenson, O. (2004). Science as a map in technological search. *Strategic Management Journal*, 25(8-9):909–928.

Hall, B. H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hargadon, A. and Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4):716–749.

Henderson, R. and Cockburn, I. (1994). Measuring competence? exploring firm effects in pharmaceutical research. *RAND Journal of Economics*, 25(1):63–84.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1):9–30.

Jaffe, A. B. (1986). Technological opportunity and spillovers of r&d: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3):577–598.

Katila, R. and Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194.

Kogut, B. and Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3):383–397.

Kogut, B. and Zander, U. (1996). What firms do? coordination, identity, and learning. *Organization Science*, 7(5):502–518.

Lavie, D., Kang, J., and Rosenkopf, L. (2011). Balance within and across domains: The performance implications of exploration and exploitation in alliances. *Organization Science*, 22(6):1517–1538.

Levinthal, D. A. and March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14(S2):95–112.

Levitt, B. and March, J. G. (1988). Organizational learning. *Annual Review of Sociology*, 14:319–338.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

Rosenkopf, L. and Almeida, P. (2003a). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Almeida, P. (2003b). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306.

Schilling, M. A. and Phelps, C. C. (2007). Interfirm collaboration networks and knowledge creation: The impact of large-scale network structure on firm innovation. *Management Science*, 53(7):1113–1126.

Stuart, T. E. (2000). Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, 21(8):791–811.

Stuart, T. E. and Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38.

Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Trajtenberg, M. (1990). A penny for your quotes: Patent citations and the value of innovations. *RAND Journal of Economics*, 21(1):172–187.

Tripsas, M. and Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10-11):1147–1161.